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Universities claim to value community-engaged scholarship: So why do they discourage it?

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Abstract

With current crises of academic relevance and legitimacy, there is a need for epistemic equity inherent to community-engaged research. Scholars in science communication and science and technology studies have analyzed, advocated for, and conducted public engagement in pursuit of this goal. However, despite desires to celebrate public engagement, US academic institutions and organizations often present barriers to meaningful community-engaged research. From tenure and promotion requirements, to lack of recognition and resources, universities in the American academic landscape are not currently organized to support such work. In this article, we offer a conceptual framework to examine the complex structural dimensions of academic institutions that have systematically discouraged and devalued faculty participation in community-engaged scholarship. We outline four such structural dimensions: interrogating epistemic biases, neoliberalist tendencies, gendered norms, and colonial-racist defaults. Our goal is to illuminate processes that could inform interventions to bridge the gap between academic aspirations for community-engaged work and current actions in the academy that undermine it.

Keywords

community-engaged scholarship, public engagement, public participation in science, representation, science communication

1. Introduction

Academia has periodically undergone a variety of existential crises in response to bouts of self-reflexivity and public outcry. One manifestation of the current predicament is a crisis of relevance. Publics and scholars alike have been questioning the purpose of academia and the real-world applicability and use of academic research (Hoffman, 2016). The relevance of academia has been further obscured by a lack of transparency and accountability, to which calls for open science have emerged (Lewis, 2020; Marcinkowski et al., 2014; Stilgoe et al., 2014). Simultaneously, struggles

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for legitimacy have materialized in the form of assaults on science in the context of the "post-truth" era (Jasanoff and Simmet, 2017).

Community-engaged scholarship (CES) has emerged as a way to overcome these issues of relevance and legitimacy, to foreground questions of science-society relations, and to make academia more accountable and publicly informed. As acts of epistemic humility (Jasanoff, 2003), CES is a key pathway to increasing trust and accountability in the pursuit of knowledge. As acts of democratizing science (Irwin, 2002), engaged research is a way to increase the public's understanding of science and scientific process, and to increase the public's influence on the production of knowledge (Bonney et al., 2016). While there are external threats to the relevance and legitimacy of academic knowledge and its influence on shaping the world, for example through broader distrust in experts and expertise, one answer is to internally shift toward increased engagement with society (Renn et al., 2019), since institutions are known to shape engagement outcomes (Bickerstaff et al., 2010). Increased engagement between experts and publics has the capacity to increase trust, transparency, and accountability, which together may positively influence the legitimacy and impact of academic knowledge. Yet, scholars in the American academic landscape often face obstacles to meaningfully conducting community-engaged work. Despite academic institutions seeking to celebrate public engagement (Marcinkowski et al., 2014; Wade and Demb, 2009), academic organizations often present logistical, professional, and institutional obstacles (Grzanka and Cole, 2021). For example, considerations for tenure and promotion tend not to "make space" for CES, which is often time-consuming, resource-intensive, requires deeper trust-based relationships and is often seen to be on the fringes of what is considered disciplinarily "prestigious" (Hale, 2008; Robinson and Hawthorne, 2018).

In this article, we seek to shed light on precisely this issue—the institutional pressures that underpin the internal resistance facing scholars conducting engaged work in the US context. If CES is indeed a way to resolve the inner crises of legitimacy, external crises of relevance, and improve science-society relations, then this internal resistance to engaged work needs to be addressed and barriers ought to be removed. To remove these barriers requires a nuanced understanding of a range of institutional pressures that together result in logistical and professional impediments to conducting CES in academic contexts, particularly in the United States where such impediments are more apparent.

We posit that obstacles to CES are situated within larger existing societal power structures, interests, and norms, that are reproduced in academic institutions at organizational, departmental, and individual scales. Institutions are thus understood as the broader "taken for granted beliefs" as well as the set of norms, rules, and regulations that dictate trends, values, and cultures of individual organizations (Soneryd, 2016) and they shape how publics engage in science. What we refer to as structural pressures are the broader societal power relations that dictate these institutional relations (e.g., neoliberal, gendered and racial hierarchies), which in turn influence how organizations (and the actors within them) uphold or resist broad beliefs about the value of engaged work.[1] In analyzing a broad set of structural pressures, we provide a conceptual discussion of four relational biases, namely epistemic, neoliberal, gendered, and racialized biases. These biases are inherent predispositions that pervade in academia (though certainly not exclusive to academia) that are manifested and reproduced through institutionalized rules, reward systems, and tacit norms and expectations that we find discourage community-engaged work.

Building from this conceptual analysis, we seek to illuminate what transformations might be necessary to foster spaces and reward structures that incentivize community-engaged work. This article is thus both a conceptual framework—tying disparate threads around bias and structural pressure that operate in US academic contexts—and a policy framework to begin to think about

ways to incentivize and encourage CES at institutional and organizational scales. In what follows, we discuss the literature that examines CES, its importance, and ways it is discouraged.

2. Public, civic, community engagement in context

CES is widely used to denote the engagement and participation of publics *with* and *within* scientific pursuits across domains of service, teaching, and research (Beaulieu et al., 2018). Although each of these domains is important, we focus our attention to public engagement in research, as it often has the most barriers to overcome, in terms of funding, time, and disciplinary prestige, but with relatively high potential for social impact. Our conception of CES includes a variety of ways the public participates in scientific research, including practice-based research, participatory action research, community-based participatory research, citizen science, and co-production of knowledge. Engagement thus necessarily involves a two-way process of dialogue and collaboration between scholars and the public, in contrast to outreach, which is a unidirectional relation of knowledge distribution.

Academic engagement with various publics has been an ongoing interest of those within public understanding of science, studies of science communication and engagement, science and technology studies (STS), and wider disciplines. Scholars have studied, advocated for, and pursued public engagement with science and academia. Broadly, public engagement in research is a way to strengthen the relationship between scientific institutions and the public. Haywood and Besley (2014) conceptualize two dominant goals for community-engaged in research: promoting scientific literacy and democratizing knowledge production. An example evidencing the former goal of promoting scientific literacy is citizen science, which engages the public in data collection, processing, and community-designed research. Citizen science is evidenced to increase public knowledge and often leads to knowledge of science and scientific process (Bonney et al., 2016). Evidencing the latter is the tradition of opening up science to the public wherein publics are engaged in the design of the scientific process (Irwin, 2002). Stronger evidence for such emphasis on community engagement is drawn from Europe, where there are calls for "upstream engagement" and involvement of the public in the design of research (e.g., Responsible Research and Innovation) and oversight and transparency of research (e.g., Open Science). Such frameworks are arguably slower to institutionalize at the national scale in the United States, though there is certainly variegation in the uptake. What ties these threads are overarching goals of moral and social accountability (Besley et al., 2020) and responsiveness to societal needs (Beaulieu et al., 2018; Boyer, 1990). This responsiveness stems from the desire to increase the value of and funding for scientific research (Besley et al., 2020). While there have been renewed calls for responsiveness and relevance of research (Shavit and Silver, 2022), such calls are not new; they are recent manifestations of well-developed traditions such as action research pioneered by Kurt Lewin (1946).

There are well-developed literatures in STS examining public engagement (Delgado et al., 2011), and in public understanding of science examining the democratization of expertise and science engagement (Bonney et al., 2016; Riesch and Potter, 2014). Similarly, there is extensive literature in community engagement examining individual, departmental, and organizational characteristics that motivate CES (Beaulieu et al., 2018; Sandmann, 2008; Wade and Demb, 2009), logistical and professional obstacles (Cavallaro, 2016), and broader organizational barriers to engaged research (Stoecker, 2008). This existing work has taught us a tremendous amount about the role of the researcher in engaged work. What requires more attention, though, is the role of academic *institutions* and the structures that inform them. Our goal in this article, then, is to consider how and why US institutions resist engaged research, and how they might transform to encourage CES.

Core to our examination is an inherent contradiction to community-engaged work. As part of reasserting legitimacy and relevance, US academic institutions are often eager to celebrate and espouse a rhetoric of supporting public engagement (Marcinkowski et al., 2014; Wade and Demb, 2009). Yet, there tend to be substantial obstacles facing scholars who conduct CES (Grzanka and Cole, 2021; Rose et al., 2020), and American academic organizations are often not well suited to supporting CES. Organizations are more likely to support scholarly and theoretical impact than community or practical impact (Moore and Ward, 2010). Part of this is because traditional research-based scholarship is often perceived to be more rigorous and intellectual than service-based research (Colbeck and Wharton-Michael, 2006; O'Meara, 2002). In addition, in evaluations, community-engaged work is often relegated to the undervalued category of "service" (Alperin et al., 2019). Evaluation policies frequently separate research, teaching, and service, and in a way that prioritizes research and/or teaching (depending on the university focus) over service (Jaeger et al., 2012; Moore and Ward, 2010). Scholars who integrate service into their research and teaching often face career difficulties (Colbeck and Wharton-Michael, 2006), and service is seen as a distraction from traditional research (Moore and Ward, 2010). Those "conducting action research, community-based research, or applied research have been told that their research does not "count" in any significant way and that they should abandon such lines of research if they want to achieve tenure" (DeWine, 2005). This illustrates the tacit socialization of faculty in career preparation (Sandmann et al., 2008).

Generally, institutions in the American academic context tend to not only lack support but also discourage CES, though there are important exceptions and fast-evolving progressions. Alternative models do exist, with some American universities shifting toward institutionalized community-based research-practice partnerships (Robinson and Hawthorne, 2018), co-learning partnership models, such as at Tufts University's Department of Urban and Environmental Policy and Planning (Loh, 2016), and interest in knowledge co-production across some US scientific institutions (Editorial, 2018). Funding agencies have shifted in the past two decades toward seeking "broader impact" statements (e.g., the National Science Foundation), though measures of broader impact are unevenly prioritized across disciplines (Roberts, 2009) and broader impact is still seen as separate and sometimes additional to intellectual merit. Finally, citizen science, involving the public in scientific data collection, has an established history in the United States (Noss, 2020). While citizen science tends to engage and educate the public on scientific processes, publics tend to be less involved with the design of the research and the topics of data collection, and the publics that engage with citizen-science have historically not been diverse (predominantly White, affluent, highly educated) (Pandya, 2012).

In addition, while discouragement and devaluation of CES is more common in US contexts, this is not necessarily the case in other geographic contexts where there are more well-established and advanced models across a spectrum of engaged research. Publics are being invited at earlier stages of research to inform the design and problem formation in scientific research in Europe, for example, upstream engagement, science shops, and others (Leydesdorff and Ward, 2005; Pidgeon and Rogers-Hayden, 2007). Similarly, though not without critique (Hartley et al., 2017), there are efforts across Europe to institutionalize public impact and relevance of scientific research, for example through the Responsible Research Innovation (RRI), frameworks that emphasize connections between social relevance and scientific research (Owen et al., 2013). Well-established efforts also exist of strengthening integration of Indigenous knowledges into scientific research, for example, in New Zealand.

Yet, while there are forms of community-engaged research and developments in this direction, this does not mean that CES is institutionally valued and regarded as intellectually rigorous by academia. Norms and tacit expectations that deprioritize CES continue to pervade in the majority

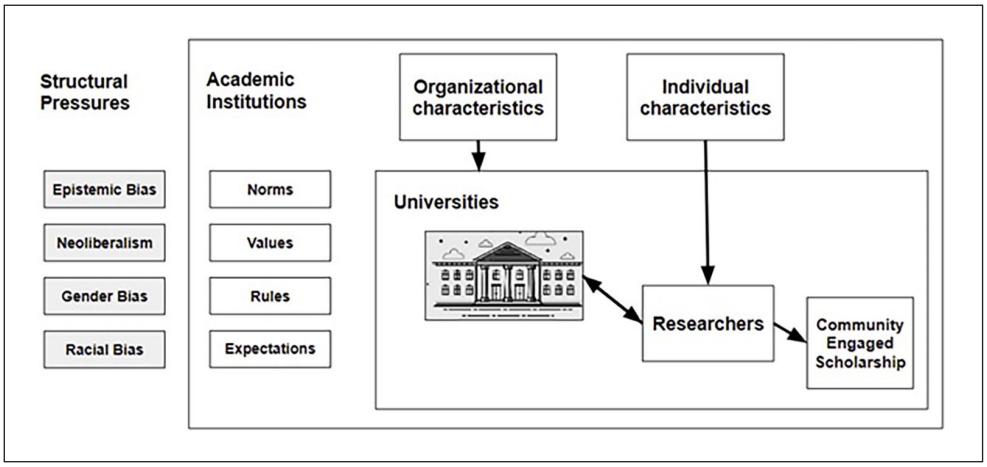


Figure 1. Conceptual framework of institutional pressures influencing organizational support for CES.

of US academic institutional contexts; nodding to a "deep-seated resistance to change that values and legitimizes CES " (Saltmarsh et al., 2009: 3).

Building from existing research that demonstrates the challenges and obstacles scholars face in conducting CES, we examine the structural underpinnings of barriers to CES in US academic institutions and organizations. Our goal is to examine why, despite desires to highlight CES, do organizations not support CES? To answer this, we offer a conceptual framework through which to interrogate the mutually reinforcing societal power structures and pressures exerted *upon* and *within* US academic institutions that influence and particularly inhibit faculty participation in CES. In what follows, we analyze four structural societal dimensions that discourage CES by exerting pressure on and within US academic institutions to discourage CES in various ways (see Figure 1). Previous conceptual models have centered the researcher (e.g., Beaulieu et al., 2018), we instead focus on the organization as a site for potential transformation, while recognizing that structural pressures both exert external influence and are internally reproduced at various institutional, organizational, and individual levels. We begin by interrogating the epistemic issues embedded within academic institutions, then moving to the neoliberalist tendencies, followed by a discussion of gendered norms, and finally, we discuss of colonial biases in academia.

3. Epistemic biases of knowledge production

Understanding the core of resistance to CES requires examination of the epistemic assumptions embedded within US academia. Academia relies on "a particular social relation of knowledge production that preserves the power of the academy to maintain control over knowledge production" (Gaventa and Cornwall, 2008; Stoecker, 2008). Control of knowledge production arguably maintains relations of power, and what counts as knowledge is "not just an epistemic act, but also, and crucially, a political one" (Foucault, 2008; Haraway, 1990:193; Pereira, 2017) as knowledge shapes the world (Harding, 2016; Mitchell, 2002). Academic institutions not only produce but also grant legitimacy to knowledge (Kajner, 2015). Consequently, there is considerable gate-keeping to preserve this locus of control, evident through metaphors of the ivory tower and academic elitism in the United States (Bok, 1982), and subsequent resistance to engaged research. That academia might have to give away knowledge is one thing, that it might have to give the means of its production (for purposes of re-legitimizing academic knowledge) is far more radical.

Indeed, outreach, often conceptualized as distributing knowledge to the public, has been historically preferred over engagement, with faculty more likely to conduct outreach than engagement (Demb and Wade, 2012) and with institutions developing outreach models centuries before emphasizing engagement (Weerts and Sandmann, 2008). This preference potentially stems from the fact that outreach does less to disrupt the power relations of knowledge production, allowing for the institution to retain power over what knowledge is produced and conveyed (Jaeger et al., 2012; Weerts and Sandmann, 2008). Outreach maintains a unidirectional approach of technocratic, hierarchical, and elitist relations that privilege academic expertise, maintain expert control over knowledge (Jaeger et al., 2012), and devalue lay/community expertise (Loh, 2016). For example, historically in the United States, land-grant institutions, in exchange for public land endowments, pioneered an emphasis on outreach centering on translating knowledge to solve real-world issues, without much community input on what issues to focus on (Weerts and Sandmann, 2008). Despite being unidirectional, such engagement was deemed transformative in the 1990s, in the context of Ernest Boyer's emphasis on "making knowledge useful" (Boyer, 1990: 28). In contrast, bidirectional and democratic relations of knowledge production were perceived as a threat to academia based on the intention to integrate lay/community knowledge into collaboratively designed research (Jaeger et al., 2012). Such work could enable communities to "challenge the knowledge control of the higher education institution, [and] compete for funding to create knowledge" (Stoecker, 2008: 52), threatening academia's perceived hold on knowledge production.

Constructed dichotomies of knowledge production

Part of the inner resistance to CES in US academia, we argue, stems from constructed divides between science and society and ideas of what constitutes proper, prestigious, or "good" science (Lewis, 2021b). These constructed dichotomies—individual/collective, objective/subjective, universal/context-based, expert/lay—are dialectical in nature, since higher education often relies on this boundary-setting to legitimate its elevated status as a legitimizer of knowledge.

Although views of knowledge tend to exist on a spectrum from the individual to the social, in higher education, knowledge is often seen to reside in the individual rather than the social (Hood, 2002; Weerts and Sandmann, 2008) and knowledge production tends to be rewarded individually. Yet, scientific endeavors must importantly remain within the norms of the collective understandings of what science is, lest the truth of scientific outcome be obscured by personal beliefs or subjectivities (Hale, 2008).

Knowledge production tends to be governed by rules of rationality, rigor, and objectivity, in opposition to community knowledge-seeking that is deemed irrational, political, and subjective. Experiments of democratizing expertise thus threaten the position of scientific expertise by putting into question the perceived objectivity of such knowledge, though many would problematize this objectivity noting how politics and science are mutually reinforcing (Felt, 2016; Haraway, 1990; Jasanoff, 2004). These differing epistemological approaches tend to indicate whether a scholar is likely to conduct CES, for example, scholars tending toward subjectivity are more likely to conduct CES (Colbeck and Wharton-Michael, 2006).

Scientific norms such as generalizability create constructed divides between universality and context-based knowledge, with preference for the universality of knowledge, in opposition to context-based community knowledge. Postcolonial STS scholars note that this divide likely emanates from colonial thinking equating Western science with universality and modernity, in contrast with the "backward, uncivilized" local knowledges of colonized nations (Philip, 2004). With pressures to produce generalizable universal research, situated, context-driven, and community-driven work is often rendered marginal to what constitutes typically revered forms of valued scholarship. There

is considerable disciplinary variation, as there have been changes over time, with the value of context becoming apparent (Bryan et al., 2021). Recent work has emphasized that often "universal" scientific claims have historically been informed by non-universal contexts, focused on overwhelmingly White, Western, Educated, Industrialized, Rich, and Democratic societies (WEIRD) (Henrich et al., 2010), putting into question the universality of such claims (Lewis, 2021b). Such developments problematize the assumptions informing scientific research, the biases toward universality, which help explain resistance to CES that is often context-based.

Questions of knowledge production also center on who decides what knowledge is legitimate, acceptable, or rigorous. Constructed expert/lay divides have historically separated the knowledgeable expert from the uninformed public. Community-based knowledge is thus relegated as non-expert and non-scientific. There have been considerable constructivist shifts toward more democratic models of knowledge that value experiential, traditional, and Indigenous knowledges (Weerts and Sandmann, 2008). Scholars such as Brian Wynne and Steven Epstein were foundational in this deconstruction of expert/lay divides centering lay contributions to scientific understanding, for example, in the contexts of sheep farmers and ecological knowledge of radiation (Wynne, 1989), and lay AIDS activists constructing credibility in medical research (Epstein, 1995). Nevertheless, expert/lay divides continue to pervade US academic contexts.

Constructed binaries of knowledge production serve to establish, support, and reproduce academic elitism and scientific exceptionalism by excluding the public (Kajner, 2015). Knowledge production is inherently political and relational. To blur the scholar/society divide could threaten the towering position of academia. Despite considerable work deconstructing these dichotomies, they continue to be reproduced within academia in ways that discourage engaged research. Similarly, there is a risk that institutions use CES to "reaffirm their dominance" by co-creating (and co-opting) community knowledge (Kajner, 2015); universities' "embracing of community engagement not only offers a powerful rhetorical device for fundraising, it also solidifies their role in the validation and legitimation of knowledge" (Giridharadas, 2019; Kajner, 2015: 164).

4. Neoliberalization of the academy

Institutional resistance to CES can also be contextualized by broader trends toward neoliberalization of the academy, evident across international contexts, but particularly strong within the United States. Higher education has shifted from serving "public interest and democratic responsibility [to] the ideals of individual responsibility, competition, and efficiency" (Brackmann, 2015). This individualism is in opposition to the pro-social values that often motivate CES (Antonio et al., 2000; O'Meara, 2002). Broadly, *academic capitalism*, or the pressure for academic organizations (and faculty) to act entrepreneurially, forces investment into "productive" and profitable activities that gain the most return, such as grant-writing, patent-production, and sometimes teaching (where students are seen as revenue streams) (O'Meara, 2008; Slaughter and Rhoades, 2000). This decreases the funding, resources, and time for less profitable activities including those that might involve communities, thereby increasing the burden for communities to bear the costs of engaged work (Loh, 2016).

The shift in priorities leads to differences in how success is measured. Trends of neoliberalization have been accompanied by emphasis on productivity and metrics, such as university/departmental rankings, citation indices (e.g., H-index), and funding measures. While this is particularly true for American academic contexts (Mountz et al., 2015), there is considerable scholarship on "metric assemblages" in Europe (Burrows, 2012; Hicks et al., 2015). Scholarly impact tends to be measured through academic research outputs such as monographs, peer-reviewed articles, and

conference presentations (Alperin et al., 2019), whereas those outlets may not value CES, and CES might rely on non-traditional outputs (e.g., policy briefs and white papers).

These metrics in turn influence evaluation policies, such as for tenure and promotion. Despite evidence to support the rigorous, systematic, and diligent nature of CES (Warren et al., 2018), traditional research is perceived to be more rigorous than community-based research, as it "requires additional intellectual ability and thus should be evaluated by higher standards than service scholarship" (Colbeck and Wharton-Michael, 2006; O'Meara, 2002). Issues with evaluation stem from the way policies are written in vague and uncertain terms and how these policies are implemented, including a lack of knowledge for how to evaluate impact (Changfoot et al., 2020; Moore and Ward, 2010; O'Meara, 2002). Policies often separate research, teaching, and service, which means engaged teaching, and engaged research often fall in the undervalued domain of service, yet when scholars integrate these dimensions in tenure dossiers, they face difficulties in career progression (Colbeck and Wharton-Michael, 2006; Wade and Demb, 2009).

The neoliberalizing trends within US academia have included an increasing precariousness and uncertainty of academic labor, including fewer faculty positions, increased contract and temporary positions, and more administrative control over academic activities (Bauder, 2005). These conditions tend to favor those with more financial independence (and thus higher socio-economic status). Such trends are a source of resistance to CES in two ways: they limit the time and resources to engage in CES, and they limit hiring and retention of scholars more likely to conduct CES. CES tends to require longer timelines and often produces slower academic output. This is not to say that community-engaged work cannot be done quickly; however, there is a tendency for community-based work, especially that which is rigorous and ethical, to require longer timeframes. This is particularly true for more integrated and involved projects where communities are part of the project design and implementation. Where the neoliberal academy emphasizes productivity and industriousness, work that is slow can be considered unproductive or "lazy" (Mountz et al., 2015). Neoliberalization of the academy also has gendered and racialized impacts including "consistently lower proportions of marginalized people holding tenure positions or taking longer to secure tenure within many institutions" (Changfoot et al., 2020). Given the evidence that women, faculty of color, and other marginalized scholars are more likely to engage in CES, lower hiring and retention of women and faculty of color may decrease engagement in CES (Antonio et al., 2000; Settles et al., 2020). We turn next to a more detailed discussion of the gendered aspects of resistance to CES.

5. Gendered norms and their effect on the US academy

Embedded within academia are gendered biases that shape resistance to CES, including how CES is perceived and rewarded. Inherent to constructions of what is considered academically rigorous scholarship are constructed masculinist tendencies that influence resistance in two ways: work categorized as more masculinist tends to be more rewarded, and women tend to be socialized into work with more feminine tendencies, leading to devaluation and potential discrimination of women in academic contexts. With women discouraged from academia in subtle ways, and since women are more likely to engage in CES, this increases the barriers to engaged research.

Traditional models of academic knowledge production often rely on masculinist ideals that permeate academia and have direct and indirect effects on engaged scholarship. Cheryan and Markus (2020) describe masculine defaults as "a form of bias in which characteristics and behaviors associated with the male gender role are valued, rewarded, or regarded as standard, normal, neutral, or necessary aspects of a given cultural context." While these types of constructed defaults "often do not appear to discriminate by gender . . . [they] result in disadvantaging more women

than men" (Markus, 2020: 3). While simplified to some extent, default values can be mapped onto masculine and feminine stereotypes. Masculine defaults tend to include being *agentic, self-reliant, competitive, analytical, independent, individual, risk-taker*, and *context-independent*, whereas feminine defaults include being *communal, nurturing, collaborative, holistic, interdependent, relational, patient*, and *context-dependent.* These defaults can then correspond to notions of traditional and engaged scholarship and epistemology.

While scientific pursuits are often collaborative, knowledge and its production tend to traditionally reward more agentic, self-reliant, competitive, individual, and separate endeavors (Cheryan and Markus, 2020). The universalist tendencies of much academic work seek to generalize to hold epistemic legitimacy (Lewis, 2021b). Community-engaged work, on the contrary, tends to be more interdependent, communal, patient, context-dependent, and requiring nurtured collaborative relationships. If masculine defaults that align with traditional scholarship tend to be rewarded and held as the norm, then engaged work will often be viewed as deviant, less rigorous and less rewarded (O'Meara, 2002), offering some explanations for why CES is less valued in US academic contexts.

The second layer of gendered effects relates to general discrimination of women in US academic contexts. Masculine defaults in professional and academic settings tend to favor scholars that adhere to masculine gender roles, leading to less recognition, reward, and value of women, resulting in an underrepresentation of women in many male-dominated settings, including academic contexts (Cheryan and Markus, 2020). These qualities that explain underrepresentation of women in some academic contexts may also explain underrepresentation of faculty of color, women, and queer faculty. Gender and race appear to be significantly associated with collaborative and community-engaged work. For example, 90% of faculty who self-identified as being involved in service scholarship were women, and 35% were faculty of color (O'Meara, 2008). Women are more likely to "use their scholarship to address community needs" and "collaborate with local community in research/teaching," similarly faculty of color are more likely (57%) to use their scholarship to address community needs than their White counterparts (Vogelgesang et al., 2010). Epistemological approaches to academic scholarship tend to also drive CES, and women and faculty of color tend to have epistemological groundings more compatible with CES (Ward, 2010). With women faculty, faculty of color, and queer faculty more likely to engage in work that does not epitomize masculine defaults, they are less likely to be rewarded and valued leading to less promotion. This can mean shifting to work that is more traditional (leading to less CES work), or to lower faculty retention due to biases both against the scholar and their engaged work (Settles et al., 2020).

6. Colonial origins and racialized aspects of US academia

Examining the colonial aspects of American academic contexts arguably merges the epistemic, neoliberal, and gendered biases with analysis of racism, ethnocentrism, and anti-Indigeneity. Knowing that knowledge and norms are co-produced, knowledge production is understood as central to "project of the settler state. . . organized around a trope of modernity versus savagery, including bringing Christianity, capitalism, agriculture, nuclear family, and so-called reasoned inquiry" to the colonized world (Subramaniam et al., 2016: 419). Western scientific knowledge and modern science are understood to have "been co-constituted with colonialism" (Seth, 2009). These premises help explain the epistemic biases and power relations that pervade academia, which act to marginalize local, Traditional and Indigenous knowledges, produce extractive epistemic relations, impose values and systems on colonized and marginalized peoples, and prioritize the objectives of White, Western, Industrialized, Rich, and Democratic societies (Slattery et al., 2021). Contrary to this, community-engaged work, particularly that which centers communities

in knowledge production, sits contrary to those goals to emphasize local knowledges and integrate, collaborate with, and benefit communities.

These entrenched colonial academic biases shape norms and priorities in academic knowledge production, which have historically prioritized perspectives of and benefits for predominantly White and (white and male scholars and benefactors) male scholars and social benefactors. One consequence is the problematic assumptions of universality that have been recently challenged to recognize that knowledge produced within a White-dominated paradigm does not necessarily apply universally across local contexts (Lewis, 2021b). A second interconnected result of White colonial defaults of universality, that focus on Western, Educated, Industrialized, Rich, and Democratic (WEIRD) benefactors, is the marginalization of scholars of color and the type of work they are more likely to do. Faculty of color have a long history of engaged scholarship, doing work in communities not traditionally studied, and merging scholarship with public policy (Antonio et al., 2000:: 598; da Cruz, 2018) and evidence suggests that faculty of color are more likely to engage in community-engaged work (Antonio, 2002; Changfoot et al., 2020; O'Meara, 2002; Vogelgesang et al., 2010). The fact that academia has long struggled with inclusion and retention of faculty of color (Antonio, 2002), exacerbated by academic corporatization resulting in fewer faculty of color holding permanent academic positions (Changfoot et al., 2020), leads to fewer faculty conducting engaged work.

The colonial and extractive underpinnings of academia further entrench this divide. Problematically, academic knowledge has often been either extracted from communities for wider academic purpose without direct community benefit, or is selectively imparted upon communities, without recognition of communities own embedded knowledge nor of the needs of the community (Humphreys et al., 2021; Subramaniam et al., 2016). Well known is the "helicopter" model of research, also known as "safari research," "parachute research," or "neo-colonial research." This type of work typically involves more affluent and privileged scholars going into less affluent, less privileged communities to collect data, extract local knowledge without involving local knowledge holders, without benefiting the community, and without interacting with the community again (van Groenigen and Stoof, 2020). This research also often involves scholars in the global North conducting research on communities in the Global South or settler scholars doing research on Indigenous communities (Cherokee Nation, 2021). It is disrespectful to communities, fosters distrust, and exhausts communities who then see no use or relevance for academic research (Adame, 2021).

Outreach is in theory intended to move beyond helicopter research by applying knowledge to real-world problems and continuing interactions with the community through sharing and distributing the knowledge produced. Yet, such outreach activities are often unidirectional and top-down in nature, they do not necessarily recognize local knowledges, nor tailor to community needs, thus reproducing academic epistemic elitism and scholar/society divides. More collaborative and bidirectional engaged research is seen to challenge the efficient, extractive, and transactional nature of knowledge production, which some have argued to be a decolonizing of CES (Kajner, 2015).

Recognizing the colonial, White-dominated histories that have influenced scientific norms helps illuminate the biases imprinted upon academic relations that serve to create an environment in which meaningful community engagement is not adequately supported. It highlights why diversity is necessary within the academy and how lack of diversity can contribute to a lack of social relevance. In the following section, we highlight the importance of transformation and begin to map out key pathways for such transformation to occur.

7. A framework for transformation

Our objective goes beyond examining how institutions resist CES to highlight how institutions might transform. For this transformation, cultural and practical changes must be instituted. In what

follows, we develop a framework that tackles inner resistance to CES, with a focus on institutional-level change. Our four recommendations center on epistemic equity, modification of metrics, diversification of the academy, and institutional innovation.

Epistemic equity

The epistemic hierarchies academia is prone to tend to elevate academic knowledge production above other forms of knowledge production, in part resulting in the current crisis of academic relevance and legitimacy. We propose epistemic equity, or the elevating of diverse and marginalized knowledges, as one way to reconfigure epistemic relations (Bell, 2022; Kajner, 2015).

Epistemic equity first requires *reflexivity*, or critical examination, self-awareness and recognition of the epistemic bias and assumptions underpinning knowledge production and the researcher's role therein, to reflect on how we produce the knowledge that we do (Lewis and Wai, 2021). This reflexivity would be a type of institutional reflexivity, called for by scholars such as Wynne (2006), whereby an institution reflects on their role in the problem, rather than externalizing the issue to the fault of publics. Second, epistemic equity requires *humility*, the notion that expertise may be questionable, especially in conditions of scientific uncertainty (Jasanoff, 2003), coupled with entertaining the idea the knowledge scholars produce might be useless and irrelevant (Grzanka and Cole, 2021). Finally, and perhaps most difficult, is institutionalizing *recognition* and respect for lay knowledges. As Jasanoff explains in a call for humility, "the problem, of course, is how to institutionalize polycentric, interactive, and multipartite processes of knowledge-making within institutions that have worked for decades at keeping expert knowledge away from the vagaries of populism and politics" (Jasanoff, 2003: 235).

These elements of epistemic equity—reflexivity, humility, and recognition—underpin the democratization of knowledge that would enable and be enabled by engaged research and public participation in knowledge production. While challenging, such efforts would likely strengthen the knowledge produced and its relevance, leading to strengthened scholar-society relations.

Reimagining measures of success

Across disciplines, there are movements that seek incremental changes to modify measures of success in ways that encourage engaged research. A revisiting of Boyer's model for engaged scholarship reveals an argument for decoupling engaged scholarship from the domain of service (Boyer, 1990; Sandmann, 2008). While some progress has been made, recent assessments indicate more work is needed here (Alperin et al., 2019). Recent movements have seen faculty resist the fast-paced efficiency and productivity emphasis of the neoliberal academy, instead advocating for slow scholarship (Stengers, 2018), defined by quality and relevance of scientific pursuits. While the call for slow scholarship is not limited or unique to conducting engaged research, the emphasis on slower timelines could impact engaged work in meaningful ways by allowing for the relationships upon which CES is often predicated. These types of movements also tend to emphasize "community engagement and the pursuit of personally and politically meaningful work" (Mountz et al., 2015). Concretely, impact and "productivity" might need to be measured differently, either through additional metrics for engaged and societal impact, such as ways of recognizing non-academic publications, or other ways of evaluating community impact of research (Ledgerwood et al., 2021).

Diversifying the academy

In terms of countering the colonialist, racist, and sexist tendencies of academia, there is a need for structural elevation of underrepresented scholars along with cultural change to shift the norms and

expectations across academia. Given the focus of underrepresented scholars on CES, the goals of transforming institutions to integrate community-based work may align with broader recognitions of the failures of diversification in the academy (Settles et al., 2020) and thus build on existing movements toward diversification. Scholars have warned that "as long as service activities are practiced by marginalized faculty, they will remain marginalized in academe" (Antonio et al., 2000; O'Meara, 2002: 76), which only serves to reinforce the notion that structural change must be accompanied by cultural change.

The problem becomes that while we are aware of systemic bias, addressing bias in a system is notoriously difficult, for example, even when direct implicit bias training is instituted it often yields no systemic change (Dobbin and Kalev, 2018; Onyeador et al., 2021). Efforts to hire more diverse faculty are unlikely to yield long-term institutional change without addressing the underlying institutional and interpersonal biases (Momennejad et al., 2019). Thus, a focus on hiring underrepresented scholars must be accompanied by attempts to shift norms and cultures requiring broad-based institutional change (Ledgerwood et al., 2021).

Institutional innovations and reconfigurations

Institutional level priorities and configurations can significantly influence the extent to which CES is encouraged, rewarded, and valued. There are innovative examples of changes involving new models of co-learning and co-production that center engaged research at the departmental level. For example, Tufts Department of Urban and Environmental Policy and Planning developed a long-term, context-based, collaborative partnership with a community organization to overcome challenges and resistance to engaged work (Loh, 2016). In other examples, attempts to change organizational norms and restructure tenure and promotion evaluations have successfully codified support for community-engaged research (Robinson and Hawthorne, 2018). Strategies here included codifying expectations for tenure and promotion, inviting community partners to evaluations, supporting service-learning, and providing funding and resources to support engaged research. There is space for these types of departmental level changes to be scaled up at institutional levels, for example, those that influence, shape, and dictate the types of research valued, including funding agencies, professional organizations, and prestigious journals. On a broader scale, scholars such as Jasanoff have noted the "deep democratic deficit" inherent to scientific institutions, which requires engagement and reconfiguration (Jasanoff, 2016).

In democratizing knowledge production, and countering epistemic elitism and neoliberal individualism, attention could be given to revisiting the missions of universities. There are institutions well poised to lead the way in contesting the individualist productivity-focused mantras through emphasis on public service, social change, and mobilization of education for the greater good. Public institutions and land-grant institutions with publicly minded missions, in particular, would be well served in this capacity. Since institutional type and mission are argued to be the strongest indicator of public engagement (Vogelgesang et al., 2010; Ward, 2003), and faculty affiliation with a land-grant or public university strongly motivate public engagement (Peters et al., 2005; Vogelgesang et al., 2010), land-grant institutions offer transformative potential.

Attention to institutional configurations will necessarily involves institutional reflexivity to reflect on how such engagement is encouraged. As we have demonstrated, resistance to community-engaged work is deeply entrenched in ways of thinking in academia. Without institutional reflexivity, institutional innovations and reconfigurations encouraging engagement may simply reinforce existing norms and practices (Rothstein, 2013). To avoid engagement that merely furthers existing hierarchies and norms, Smith et al. (2021) discuss the operationalization of engagement practices by calling for reflexivity over what types of engagement are called for, how they are

implemented, and what outcomes of engagement are desired, before engagement practices are put into place.

8. Conclusion

In the context of academic crises of legitimacy and relevance along with increasingly complex socio-environmental challenges and injustices, there is a need for knowledge to be useful and relevant. One mechanism to achieve this is to conduct engaged research. We have presented evidence for how academics face barriers to conducting engaged work, followed by a conceptual framework to illuminate the structural forces that contribute to this resistance. This serves to recognize the role of underlying biases influencing knowledge production, from epistemic biases, neoliberal ideologies, and gendered and racialized biases that influence who is hired, the type of work they do, and how they do it. This work thus contributes to the broader fields of public understanding of science, participation in science, public and civic engagement, and science and technology studies.

Having identified a range of structural pressures, we then offer a framework for transformation. In our framework, by focusing on the institution we illuminate how larger structural initiatives can help motivate engaged work that is meaningful and impactful. Individual-centered changes are important, but without structural change, they are unlikely to shift paradigms, nor have an institutional impact (Eckel et al., 1998; Lewis, 2021a). In taking steps toward transformation, we provide four recommendations, centered on the need for epistemic equity, reimagining measures of success, diversification, and institutional innovation. Within these, we continue calls for institutional reflexivity, we suggest increased attention to the ways of thinking and doing within academic institutions, and we posit that moves toward engagement need to be designed and implemented thoughtfully to produce new norms rather than entrench existing ones.

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Note

1. Although admittedly, especially within the engagement literature, there is slippage between terms such as institution and organization, which are often used synonymously, that is, "institution" often refers to individual organizations.

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